MIL-S-55203A(MU)
AMENDMENT 2
25 July 1967
SUPERSEDING
AMENDMENT 1
18 January 1967

MILITARY SPECIFICATION

SUBASSEMBLIES, PLUG-IN, FOR

TEST SET, TRANSISTOR: AN/USM - 136

This amendment forms a part of Military Specification MIL-S-55203A(MU), dated 28 April 1966.

Paragraph 3.6.4, third sentence: Delete "plus or minus 0.01 millisecond" and substitute "plus or minus 0.02 millisecond".

Paragraph 3.6.5, second sentence: Delete "plus or minus 0.01 millisecond" and substitute "plus or minus 0.02 millisecond".

Paragraph 4.6.3(b): Delete "0.2 volts" and substitute "0.3 volt".

Paragraph 4.6.3(c): Delete "0.04 milliseconds plus or minus 0.01 milliseconds" and substitute "0.10 millisecond plus or minus 0.03 millisecond".

Paragraph 4.6.3(d): Delete "two (2)" and substitute "five (5)".

Paragraph 4.6.3(e): Delete "ten (10)" and substitute "eleven (11)".

Paragraph 4.6.3(f): Delete "0.2 milliseconds plus or minus 0.01 milliseconds" and substitute "0.22 millisecond plus or minus 0.04 millisecond".

Paragraph 4.6.3(g): Delete "5 volts amplitude at a pulse width of 1.96 milliseconds" and substitute "3 volts amplitude at a pulse width of 1.84 milliseconds".

Paragraph 4.6.3(h): Delete "98 pulses" and substitute "92 pulses".

Paragraph 4.6.4(b): Delete "0.2 volts" and substitute "0.3 volt".

Paragraph 4.6.4(c): Delete "0.04 milliseconds plus or minus 0.01 millisecond" and substitute "0.10 millisecond plus or minus 0.03 millisecond".

Paragraph 4.6.4(d): Delete second sentence.

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Paragraph 4.6.4(f): Delete "1.96 milliseconds" and substitute "1.84 milliseconds". Delete last sentence.

Paragraph 4.6.7: Delete and substitute the following:

"4.6.7 Scaling amplifier.- Testing the scaling amplifier to meet the requirements of 3.6.8 shall be done in the following manner:

(a) Insert the plug-in subassembly, Scaling Amplifier (A900), into the test circuit of Figure 4 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period.

(b) Ground pin 12 and pin 9. Looking at Test Point TP901 (collector of Q909), adjust Tailor Resistor R917 so that voltage on TP901 is approximately minus 12 volts. Remove grounds from pins 9 and 12. Now adjust R901 (BALANCE) until same minus 12 volts appears on TP901 (collector of Q909).

(c) Place the test switch SW-1 in the X1000 position, (connect pin 19 to pin 24 and pin 22 to pin 3 or energize relays K901 and K902 for gain equals X1000). Place a 10 millivolt peak-to-peak square wave into pin 7, the input of the scaling amplifier. Observe with an oscilloscope Test Point TP901 (collector of Q909), adjust R927 (X1000 ADJ.) for a 10 volt peak-to-peak square wave.

(d) With same set-up as in step c, change the test switch SW-1 to X100, (connect pin 1 to pin 5 or energize Relay K902 only for gain equals X100). Raise input level to 0.1 volt peak-to-peak square wave. Adjust R911 (X100 ADJ.) for 10 volts peak-to-peak square wave or TP901.

(e) Change the test switch SW-1 to the X10 position, (disconnect pin 22 from pin 3, pin 1 from pin 5, leave connected pin 19 to pin 24 and also connect pin 3 to pin 23 (or energize Relay K901 only for gain equals X10). Raise input to 1 volt peak-to-peak square wave and adjust R928 (X10 ADJ.) for 10 volts peak-to-peak square wave on Test Point TP901 (collector of Q909).

(f) With the test switch SW-1 in the X1 position, apply an input signal of 9.8 volts 60 cycles per second square wave signal to the input capacitor C1 (which is connected to pin 7 of J901). The output DC voltage measure by a 10 volt voltmeter having an accuracy of plus or minus 0.5 percent connected to pin 21 of J901 shall read plus 9.8 volts plus or minus 2 percent.

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(g) A 1 megohm plus or minus 5 percent 1/2 watt test resistor shall be connected in series between the source of input voltage and the input capacitor C1. Repeat test (f) above and note that the voltage read on the DC output voltmeter shall be greater than half of the value measured during test (f).

(h) Remove the 1 megohm test resistor and reconnect the input signal to the input capacitor C1. Change the input signal to 0.98 volts 60 cycles per second square wave and the test switch SW-1 to the X10 position. Observe that the DC output voltmeter reading is plus 9.8 volts plus or minus 2 percent.

(i) Change the input signal to 0.098 volts, 60 cycles per second square wave and the test switch SW-1 to the X100 position and observe that the DC output voltmeter reading is plus 9.8 volts plus or minus 2 percent.

(j) Change the input signal to 0.0098 volts 60 cycles per second square wave and the test switch SW-1 to the X1000 position and observe that the DC output voltmeter reading is plus 9.8 volts plus or minus 2 percent."

Figure 4: The value of the undesignated capacitor between pin 21 and pin 14 ground shall be C_2, 1uf, 30V.

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